

AI USE CASE INTEGRATION IN TEACHING-LEARNING

Part A: AI Use Case Profile

1. **Use Case ID:** KJS-CES-02
2. **Use Case Title:** AI-Enabled Micro-channel Battery Thermal Management System for Electric Vehicles
3. **Application Domain:** Climate, Energy, & Sustainability
4. **Collaborating Organization:** Somaiya Vidyavihar University (Funded Research Project)
5. **Faculty Owner(s):**
 - Dr. Siddappa Bhusnoor, Professor, Department of Mechanical Engineering, K J Somaiya School of Engineering
 - Dr. Manoj Pawar, Professor, Department of Mechanical Engineering, K J Somaiya School of Engineering
6. **Beneficiaries:** EV manufacturers, battery pack designers, thermal engineers, battery researchers, academia, and sustainable mobility stakeholders.
7. **Problem Statement:** Lithium-ion batteries generate significant heat during charging and discharging. Excessive temperature and non-uniform thermal distribution reduce battery life, efficiency, safety, and may lead to thermal runaway. Conventional cooling systems often suffer from higher energy consumption and lower cooling effectiveness.
8. **Background and Context:** The research focuses on developing an advanced Battery Thermal Management System (BTMS) using micro-channel heat exchangers integrated with $\text{Al}_2\text{O}_3/\text{water}$ nanofluids. Numerical and experimental investigations demonstrated improved heat transfer, enhanced temperature uniformity, and reduced pumping power requirements compared with conventional cooling approaches.

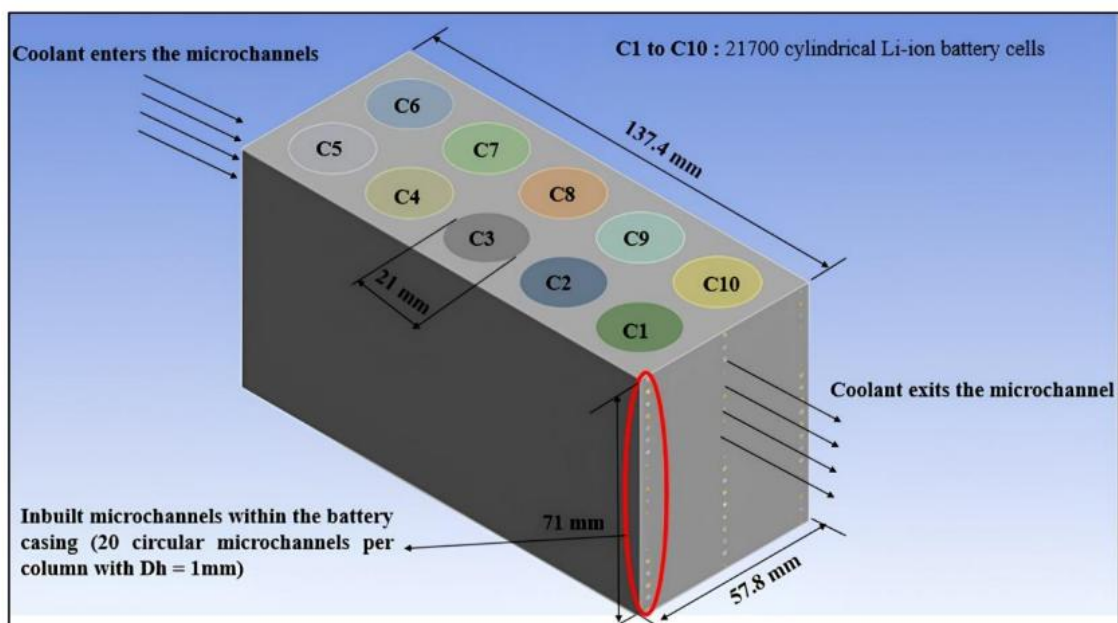


Fig. 1: Schematic representation of the battery pack geometry integrated with microchannels for battery thermal management

9. Objectives:

- 1) To optimize microchannel geometry for enhanced thermal performance.
- 2) To select and optimize nanofluid concentration.
- 3) To develop AI-assisted predictive models for thermal behaviour.
- 4) To validate numerical results through experiments.
- 5) To establish an intelligent BTMS framework for EV applications.

10. Solution Approach: Some concepts for solving this use case- Sensor data acquisition, CFD simulations, AI/ML-based prediction, optimization of coolant parameters, experimental validation, and development of a digital thermal management framework. The proposed solution focuses on the development of an AI-Enabled Micro-channel Battery Thermal Management System (BTMS) for electric vehicles by integrating advanced thermal engineering, numerical simulation, machine learning, and experimental validation. The battery pack consists of ten cylindrical 21700 lithium-ion cells integrated within an aluminium casing containing embedded circular microchannels of 1 mm hydraulic diameter. Coolant flows through these microchannels to extract heat generated during charging and discharging cycles. Experimental and numerical investigations demonstrated that microchannel-based cooling provides superior temperature control and thermal uniformity compared to conventional cooling systems.

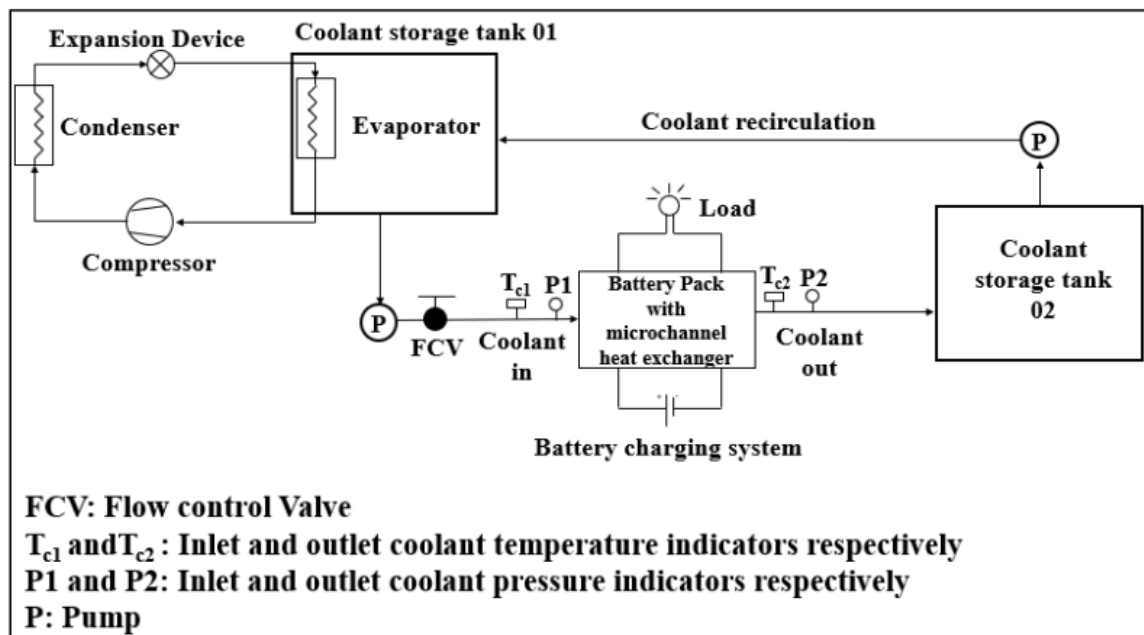


Fig. 2: Schematic of bench-scale experimental setup of the battery thermal management system with a microchannel heat exchanger

Stage 1: Battery Thermal Data Acquisition

- Installation of temperature sensors at multiple battery locations.
- Collection of real-time temperature, voltage, current, and coolant flow data.
- Creation of a thermal performance database under varying operating conditions.

Stage 2: CFD-Based Thermal Modeling

- Development of 3D battery pack and microchannel models using CAD and ANSYS Fluent.
- Simulation of heat transfer and fluid flow under different Reynolds numbers (400–700).
- Evaluation of temperature distribution, pressure drop, and heat transfer coefficient.

Stage 3: Nanofluid Optimization

- Investigation of Al_2O_3 /water nanofluids with different particle concentrations.

- Analysis of thermo-hydraulic performance and pumping power requirements.
- Identification of optimum nanofluid concentration providing maximum cooling with minimum energy consumption.

Stage 4: AI and Machine Learning Integration

- Development of predictive machine learning models using CFD and experimental datasets.
- Prediction of battery surface temperature, hotspot locations, and thermal gradients.
- AI-assisted optimization of coolant flow rate, inlet temperature, and nanoparticle concentration.

Stage 5: Digital Twin Development

- Creation of a virtual replica of the battery cooling system.
- Continuous synchronization of sensor data with simulation models.
- Real-time performance monitoring and predictive diagnostics.

Stage 6: Experimental Validation

- Fabrication of a bench-scale battery thermal management test rig.
- Validation of simulation results using real battery packs and thermal sensors.
- Continuous refinement of AI prediction models using experimental observations.

Expected Technical Outcomes

- Maximum battery temperature maintained below critical safety limits.
- Enhanced temperature uniformity ($\Delta T < 5^\circ\text{C}$).
- Reduced thermal runaway risk.
- Reduced coolant consumption.
- Improved battery life and charging performance.
- Intelligent predictive thermal management capability.

11. Operational Challenges:

- Heat Transfer vs Pressure Drop Trade-Off:** Increasing coolant flow improves cooling performance but also increases pressure losses and pumping power requirements. An optimal operating region must be identified.
- Nanofluid Stability:** Long-term stability of Al_2O_3 nanoparticles in water is essential. Particle agglomeration may reduce thermal conductivity and cause blockage within microchannels.
- Microchannel Manufacturing Precision:** The cooling performance depends heavily on microchannel dimensions. Minor fabrication deviations can significantly alter coolant flow distribution.
- Sensor Accuracy and Reliability:** Temperature sensors must operate accurately in high-temperature and vibration-prone EV environments.
- CFD Model Validation:** Simulation predictions must consistently match experimental measurements across different operating conditions.
- AI Model Generalization:** Machine Learning models trained on laboratory data should remain reliable under real-world driving conditions.
- Computational Requirements:** Real-time prediction requires lightweight AI models suitable for edge deployment within vehicle control units.
- Battery Aging Effects:** Battery thermal behavior changes with aging and repeated charge-discharge cycles. AI models must adapt accordingly.
- Integration with Battery Management Systems (BMS):** The proposed BTMS must seamlessly interact with existing BMS architectures and vehicle control systems.
- Scalability and Commercial Deployment:** The cooling architecture should be scalable from laboratory prototypes to commercial EV battery packs without substantial cost increases.

12. High-Level Workflow:

Step 1: Battery Operation

- Battery undergoes charging/discharging cycles.
- Heat generation occurs due to electrochemical reactions and internal resistance.

Step 2: Thermal Data Acquisition

- Sensors continuously measure:
 - Battery temperature
 - Coolant temperature
 - Voltage
 - Current
 - Flow rate

Step 3: Data Preprocessing

- Raw sensor data is filtered and normalized.
- Thermal anomalies are identified.

Step 4: AI-Based Thermal Prediction

- Machine learning models predict:
 - Maximum battery temperature
 - Temperature distribution
 - Potential hotspot formation
 - Cooling requirements

Step 5: CFD-Based Digital Twin Analysis

- Real-time operational data updates the virtual battery model.
- Thermal performance is simulated under current operating conditions.

Step 6: Optimization Engine

- AI determines:
 - Optimal coolant flow rate
 - Best Reynolds number
 - Nanofluid concentration
 - Cooling strategy

Step 7: Intelligent Cooling Control

- Cooling system automatically adjusts operating parameters.
- Microchannel cooling maintains uniform temperature distribution.

Step 8: Experimental Validation

- Predicted results are compared with sensor measurements.
- Models are continuously refined.

Step 9: Performance Monitoring Dashboard

- Engineers monitor:
 - Battery health
 - Cooling efficiency
 - Energy consumption
 - Safety indicators

Step 10: Decision Support and Predictive Maintenance

- AI provides recommendations for preventive maintenance.
- Potential thermal failures are predicted before occurrence.

13. Expected Outputs:

The proposed AI-Enabled Micro-channel Battery Thermal Management System (BTMS) is expected to generate significant outcomes in the areas of thermal engineering, artificial intelligence, electric vehicle technology, and interdisciplinary engineering education. The major expected outputs are described below:

a. Optimized Micro-channel Battery Thermal Management System

A fully optimized BTMS incorporating circular microchannel heat exchangers will be developed for lithium-ion battery packs. The system will demonstrate enhanced heat dissipation capability while maintaining battery temperatures within the recommended operating range (15–45°C). The optimized design will significantly reduce thermal hotspots and improve temperature uniformity across battery cells.

b. AI-Based Thermal Prediction Model

A machine learning framework will be developed using numerical & experimental datasets to:

- Predict battery surface temperatures.
- Identify potential hotspot locations.
- Estimate cooling requirements under varying operating conditions.
- Forecast thermal behavior during charging and discharging cycles.
- Support real-time thermal decision-making.

The AI model will act as an intelligent decision-support system for future Battery Management Systems (BMS).

c. Digital Twin of Battery Cooling System

A digital twin platform will be developed to create a virtual representation of the physical battery pack and cooling system. The digital twin will enable:

- Real-time monitoring.
- What-if analysis.
- Predictive maintenance.
- Performance optimization.
- Remaining useful life estimation.

This output aligns with Industry 4.0 and Smart Manufacturing initiatives.

d. CFD-Based Thermal Performance Database

A comprehensive database will be generated through ANSYS Fluent simulations containing:

- Reynolds number variations (400–700).
- Different coolant flow rates.
- Various microchannel geometries.
- Multiple inlet temperatures.
- Different nanoparticle concentrations.

The database will serve as a valuable resource for AI model training and future EV battery research.

e. Nanofluid Performance Optimization Framework

The study will establish optimum operating conditions for Al_2O_3 /water nanofluids by identifying:

- Best nanoparticle concentration.
- Maximum heat transfer coefficient.
- Minimum pumping power requirements.
- Optimum coolant flow conditions.

The framework can be directly adopted by EV battery manufacturers for advanced cooling system design.

f. Experimental Validation Test Rig

A fully functional experimental setup will be developed consisting of:

- Lithium-ion battery module.
- Microchannel cooling assembly.
- Coolant circulation system.
- Temperature sensors.
- Data acquisition system.

The setup will validate numerical and AI predictions and serve as a future research platform for students and researchers.

g. Intelligent Cooling Control Strategy

An adaptive cooling strategy will be developed where AI algorithms recommend:

- Coolant flow rate adjustments.
- Cooling activation thresholds.
- Thermal balancing strategies.
- Energy-efficient cooling operation.

This can ultimately lead to autonomous battery thermal management systems for next-generation electric vehicles.

14. Modern AI Governance and Paradigms:

Area	Relevance	Recommended Consideration
Privacy & Data Protection	Medium	Ensure secure storage and transmission of battery performance data, sensor readings, simulation results, and operational logs. Apply controlled access mechanisms for research datasets and industrial partner data.
AI Security & Trustworthiness	High	Validate AI-generated temperature predictions and optimization recommendations against CFD simulations and experimental results. Establish model accuracy thresholds and uncertainty estimation before deployment in EV applications.
Responsible AI	High	Ensure AI models provide explainable recommendations for coolant flow rate, nanofluid concentration, and thermal management strategies. Maintain transparency in model assumptions and decision-making processes.
Generative AI / LLMs	Medium	Utilize LLMs for literature review, technical documentation, automated report generation, knowledge extraction, and interpretation of simulation results while ensuring traceability and verification of outputs.
Agentic AI	Medium	Future integration of autonomous optimization agents for adaptive thermal management, parameter tuning, anomaly detection, and predictive maintenance recommendations in Battery Management Systems (BMS).
Human-in-the-Loop Decision Making	High	Thermal engineers and researchers must review AI-generated recommendations before implementation. Critical decisions affecting battery safety, cooling strategies, and design modifications should remain under expert supervision.
Regulatory / Governance Considerations	High	Ensure compliance with EV safety standards, battery testing protocols, ISO 26262 functional safety principles, and industry guidelines for battery thermal management systems. Maintain proper documentation, validation records, and experimental traceability.
Model	High	Continuously compare AI predictions with experimental

Validation & Reliability		measurements and CFD results. Periodically retrain models using new operational data to maintain prediction accuracy throughout battery life cycles.
Digital Twin Governance	Medium	Establish verification and synchronization mechanisms between the physical battery pack and its digital twin to ensure reliable real-time monitoring and predictive analytics.
Sustainability & Environmental Impact	High	Ensure AI-assisted optimization minimizes energy consumption, coolant usage, and thermal losses, thereby supporting sustainable transportation and carbon emission reduction goals.
